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Stack

Q.No:1

Problem Description

A and B are playing a game. In this game, both of them are initially provided with a list of n numbers. (Both have the same list but their own copy).

Now, they both have a different strategy to play the game. A picks the element from **start of his list**. B picks from the **end of his list**.

You need to generate the result in form of an output list.

Method to be followed at each step to build the output list is:

1. If the number picked by A is **bigger than** B then this step's **output is 1**. B **removes** the number that was picked from their list.
2. If the number picked by A is **smaller than** B then this step's **output is 2**. A **removes** the number that was picked from their list.
3. If both have the **same number** then this step's **output is 0**. Both A and B **remove** the number that was picked from their list.

This game **ends** when at least one of them has no more elements to be picked i.e. when the **list gets empty**.

Output the built output list.

Constraints

$1 \leq N \leq 10^6$

$1 \leq \text{num} \leq 10^9$

Input format:

First line consists of a number n , size of the list provided.

Next line consists of n numbers separated by space.

Output format:

Output the required output list.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
1 2 3
```

EXPECTED OUTPUT

```
2 2 0
```

Test Case 2

INPUT (STDIN)

```
5
5 7 1 2 3
```

EXPECTED OUTPUT

```
1 1 1 2 0 2 2 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
if(a[i]>a[j])
```

Code:

```
#include <stdio.h>
```

```

int main()
{
int n;
scanf("%d", &n);
int a[n];
for (int i = 0; i < n; i++)
{
scanf("%d", &a[i]);
}
int i = 0;
int j = n - 1;

while (i < n && j >= 0)
{
if (a[i] > a[j])
{
printf("1 ");
j--;
}
else if (a[i] < a[j])
{
printf("2 ");
i++;
}
else
{
printf("0 ");
i++;
j--;
}
}

return 0;
}

```

Output:

Test1:

```

5
5 7 1 2 3
1 1 1 2 0 2 2 2 %c

```

Test2:

```

3
1 2 3
2 2 0 %c

```